

ISHAN BHARDWAJ

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Quantitative finance graduate student with experience in credit risk modeling, derivatives pricing, and AI model evaluation for financial applications. Built structural credit models (Merton), options pricing engines, and risk analytics from scratch. Targeting credit risk, bank strats, quant risk, and fixed income analytics roles.

EDUCATION

University of Minnesota — Twin Cities M.S. Financial Mathematics	Expected Dec 2026
KIIT University B.Tech, Computer Science	Jul 2020 – May 2024

SKILLS

Risk & Credit	Merton structural model, Altman Z-Score, Distance-to-Default, PD estimation, VaR (historical + parametric), CVaR, KS statistic, Gini coefficient, Brier Skill Score, Basel II validation metrics
Quant & Derivatives	Black-Scholes, binomial trees, Monte Carlo, Greeks, implied volatility, delta hedging, options pricing, cointegration analysis
ML & Data	Python (pandas, numpy, scikit-learn, XGBoost), SHAP TreeExplainer, logistic regression, stratified CV, class-imbalance handling, pyarrow, SQL-style processing

WORK EXPERIENCE

Quantitative Finance Expert (Contractor)	Jul 2026 – Present
Mercor	

- Evaluate AI model outputs on graduate-level quant finance tasks — derivatives pricing, options Greeks, structural credit models (Merton), systematic trading logic — providing authoritative feedback on mathematical accuracy and reasoning quality.
- Design ground-truth benchmarks for stochastic processes, risk model construction (VaR/CVaR/Sharpe), and portfolio optimization; outputs used in training frontier AI systems.

Equity Research Analyst Intern	Jun 2025 – Aug 2025
Atlasio Capital Partners	

- Built predictive financial models across 40+ companies, identifying margin inefficiencies of 300–500 bps; designed automated data workflows reducing inconsistencies by 25%.
- Developed scenario-based pricing and cost sensitivity models across 10+ assumptions; Python pipeline automation improved research turnaround by 20%.

Quantitative Analyst Intern	Jan 2024 – Mar 2024
Mudraksh & McShaw	

- Analyzed 500K+ records identifying statistical trends and cost drivers; built automated pipelines reducing processing time by 20–25% and KPI dashboards tracking 15+ metrics.

Trade Support Analyst Intern	Nov 2023 – Dec 2023
AlgoBulls	

- Validated 1,000+ daily transactions across financial systems with near 100% accuracy; supported audit workflows across 10,000+ transactions.

PROJECTS

Credit Default Model — Merton vs Altman vs XGBoost	GitHub
Python · Merton (1974) Structural Model · Altman Z-Score · XGBoost · SHAP TreeExplainer · 28 Financial Ratio Features	

- Merton structural model (equity = call on firm assets) solved via Newton-Raphson; AUC 0.956 / KS 0.917 on 46 companies — outperforming Altman (0.892) and XGBoost (0.778). SHAP confirmed log(market cap) and equity vol as top predictors.

Options Pricing Engine — From Scratch, No QuantLib	GitHub
Python · Black-Scholes · CRR Binomial · Monte Carlo · Brent/Newton IV Solvers · Delta-Hedging Simulation	

- Three independent pricers (BS analytical, CRR binomial w/ American early exercise, MC w/ antithetic variates for European/Asian/barrier) with all 5 Greeks verified to <1e-4 via finite-difference; IV round-trip <1e-6. Delta-hedging simulation validates vol carry formula.

CME Systematic Futures Trading System — Top 10%, CME Institute Competition	GitHub
Python 3.12 · pandas · numpy · pyarrow · asyncio · IBKR API (ib_insync) · Custom backtesting engine	

- End-to-end automated futures system across 5 independent strategies on ES/NQ/CL; OOS Sharpe 4.85, IS/OOS ratio 1.34x, profitable in all 5 held-out years (2022–2026). Fill validity audit corrected win rate from 79.9% to honest 53–60%.
- Explored 15 signal types, rejected 11 after OOS failure with documented reasons; 38-check test suite verifying P&L to the cent.